

PROJECT DESIGN PHASE-II | COMPREHENSIVE TECHNICAL REPORT

Date	31 January 2025	Team ID	6ab61594fa5e5219dc387253
Project Name	Import Data using Transform Maps	Maximum Marks	2 Marks / 5 Marks

1. PROBLEM – SOLUTION FIT CANVAS ANALYSIS

The Problem–Solution Fit validates that customer pain points are accurately identified and that the proposed ServiceNow data import and transformation solution effectively resolves them.

Dimension	Detailed Analysis & Mapping
1. Customer Segment(s) [CS]	Enterprise ServiceNow administrators, IT service desk managers, HR data migration specialists, and system integrators handling bulk employee, asset, or organizational record onboarding.
2. Jobs-to-be-Done / Problems [J&P;]	• Cleanly migrating legacy CSV/Excel data into ServiceNow tables. • Preventing duplicate records during bulk data ingestion. • Mapping source columns to target fields correctly. • Validating data formats prior to insertion and maintaining referential integrity (manager, department references).
3. Triggers [TR]	• Frequent data corruption and errors during manual spreadsheet uploads. • Compliance audits flagging duplicate user or asset records. • Upcoming organizational restructuring requiring massive bulk user imports. • Encountering broken reference field errors during routine data loads.
4. Emotions: Before / After [EM]	Before: Frustrated, anxious, overwhelmed by manual data cleansing, stressed about breaking production database references, fearful of audit penalties. After: Confident, relieved, in control, assured of data accuracy, efficient, and audit-ready.
5. Available Solutions [AS]	• Manual CSV imports via standard ServiceNow list view. • Custom external ETL scripts (Python, Talend). • Basic out-of-the-box transform maps without validation scripts. Pros & Cons: Manual uploads cause duplicate entries and broken references; external ETL tools require complex middleware and separate infrastructure.
6. Customer Constraints [CC]	• Strict organizational security policies restricting external middleware. • Limited administrator time for manual spreadsheet error-checking. • Budget constraints for third-party ETL licenses. • System performance limits on large data batches.
7. Behaviour [BE]	• Spends hours manually cross-referencing spreadsheets in Excel before uploading. • Writes ad-hoc background scripts to clean data. • Manually fixes broken reference fields one by one in ServiceNow tables.
8. Channels of Behaviour [CH]	8.1 Online: ServiceNow Community forums, developer documentation, ServiceNow Academy courses, GitHub repositories for transform scripts, IT service management webinars. 8.2 Offline: Enterprise IT architecture review meetings, internal workshops on data governance, administrator training sessions.
9. Problem Root Cause [RC]	Legacy data systems lack standardized validation rules; standard out-of-the-box import sets lack contextual relational mapping and intelligent deduplication logic; manual data entry introduces human error and inconsistent formatting.
10. Your Solution [SL]	An advanced ServiceNow data ingestion and transformation framework using Import Sets and Custom Transform Maps. Features automated Coalesce keys (u_emp_id) for intelligent deduplication, custom onBefore validation scripts, dynamic dot-walking to resolve reference fields (sys_user, cmn_department), and rigorous role-based Access Control Lists (ACLs) ensuring secure, error-free bulk data onboarding.

2. AGILE SPRINT PLANNING & VELOCITY FRAMEWORK

To ensure disciplined execution, the project is structured across Agile sprints using Fibonacci story point estimation. The breakdown of Epics, User Stories, and team velocity is detailed below.

Agile Planning Definitions & Estimation Scale:

- **Sprint:** Fixed period in which a team works to complete a designated set of tasks.
- **Epic:** Large body of work broken down into smaller User Stories across multiple sprints.
- **Story Point Scale:** Very Easy (1 SP) | Normal (2 SP) | Moderate (3 SP / 4 SP) | Difficult (5 SP).

Sprint 1 Breakdown — Total: 12 Story Points

Epic	User Story / Task (USN)	Complexity / Effort	Story Points
Epic 1: Data Import	Preparing Source Data (USN1)	Normal task	2
	Creating Import Set (USN2)	Normal task	2
	Loading Source Data (USN3)	Very Easy task	1
Epic 2: Data Transformation	Creating Transform Map (USN4)	Moderate task	3
	Mapping Source and Target Fields (USN5)	Moderate task	3
	Configuring Coalesce / Duplicate Handling (USN6)	Moderate task	3

Total Sprint 1 Points = 2 + 2 + 1 + 3 + 3 + 3 = 14 SP

Sprint 2 Breakdown — Total: 16 Story Points

Epic	User Story / Task (USN)	Complexity / Effort	Story Points
Epic 3: Data Validation	Validating Imported Records (USN7)	Normal task	2
	Handling Missing / Invalid Data (USN8)	Moderate task	3
	Testing Reference Fields (USN9)	Moderate task	3
Epic 4: Deployment & Verification	Running Transform (USN10)	Moderate task	3
	Verifying Target Records (USN11)	Normal task	2
	Error Handling and Final Testing (USN12)	Moderate task	3

Total Sprint 2 Points = 2 + 3 + 3 + 3 + 2 + 3 = 16 SP

TEAM VELOCITY CALCULATION

Metric	Calculation / Formula	Description
Total Story Points Completed	14 (Sprint 1) + 16 (Sprint 2) = 30	Sum of all completed story points across sprints
Number of Sprints	2 Sprints	Total completed iteration cycles
Team Velocity	30 / 2 = 15 Story Points / Sprint	Average velocity capacity per sprint

Conclusion: The project combines Problem–Solution Fit validation with structured Agile sprint planning. The planned Import Data using Transform Maps implementation covers data import, transformation, validation, deployment and verification across two sprints, with a calculated velocity of 15 Story Points per Sprint.